

Course Name: Business Intelligence

Course Code: CCDS 313



Semester: Second Semester, 2026

Assignment 2

Business Intelligence Model Development

Saudi Superstore Sales Intelligence System (2011-2014)

Prepared By :

Student Name	Student ID
Raghad Saleh Sindi	2311173
Fawziah Abdulrahem Alamri	2311094
Jody Jamal Bokhary	2310890

Submitted To:

Dr.Roshuo Mohammed

Introduction

Case Study Overview

Our project focuses on the retail sector in Saudi Arabia, specifically the "Saudi Superstore". We chose this domain because the Saudi retail market is one of the fastest-growing in the region, offering a rich dataset for applying Business Intelligence tools.

Project Objective

The main goal is to develop a hybrid BI model (Descriptive, Predictive, and Prescriptive) to optimize sales performance across different Saudi regions like Riyadh, Jeddah, and Dammam.

Problem Statement & Objectives

The Saudi Superstore faces challenges in translating historical data into actionable strategies. While certain regions like Riyadh show high sales, there is a lack of guidance on optimizing profit and managing the volatility shown in the 2011-2014 data. Our objective is to bridge this gap using a full-spectrum BI approach:

- Descriptive: Identifying historical sales leaders in Saudi cities.
- Predictive: Forecasting future revenue with high accuracy.
- Prescriptive: Formulating data-driven decisions to maximize growth.

Part 1: Descriptive Analytics (Power BI Dashboard)

Introduction to Power BI

This project utilizes Microsoft Power BI, a leading Business Intelligence tool, to transform raw historical data into an interactive dashboard. The primary goal is to perform Descriptive Analytics to understand "what happened" in the Saudi Superstore between 2011 and 2014, allowing for data-driven decision-making.

Descriptive Question:

"What are the historical trends of sales and profit across product categories and sub-categories in Saudi Arabia's retail market over time, and which cities and customer segments contribute most to total sales and profit?"

Tool Justification

Microsoft Power BI was chosen for its superior capability in geographic visualization (Saudi Maps) and interactive slicing, which is essential for descriptive storytelling. Its drag-and-drop interface enables non-technical users to build professional dashboards rapidly.

Methodology

Data Preparation:

Data was imported and cleaned to ensure accuracy across key fields such as Sales, Profit, Category, and Geography (Saudi Cities).

Dashboard Design:

A user-centric layout was designed to balance high-level summaries with granular details, utilizing Slicers for dynamic filtering by Date, City, and Category.

Dashboard Screenshots

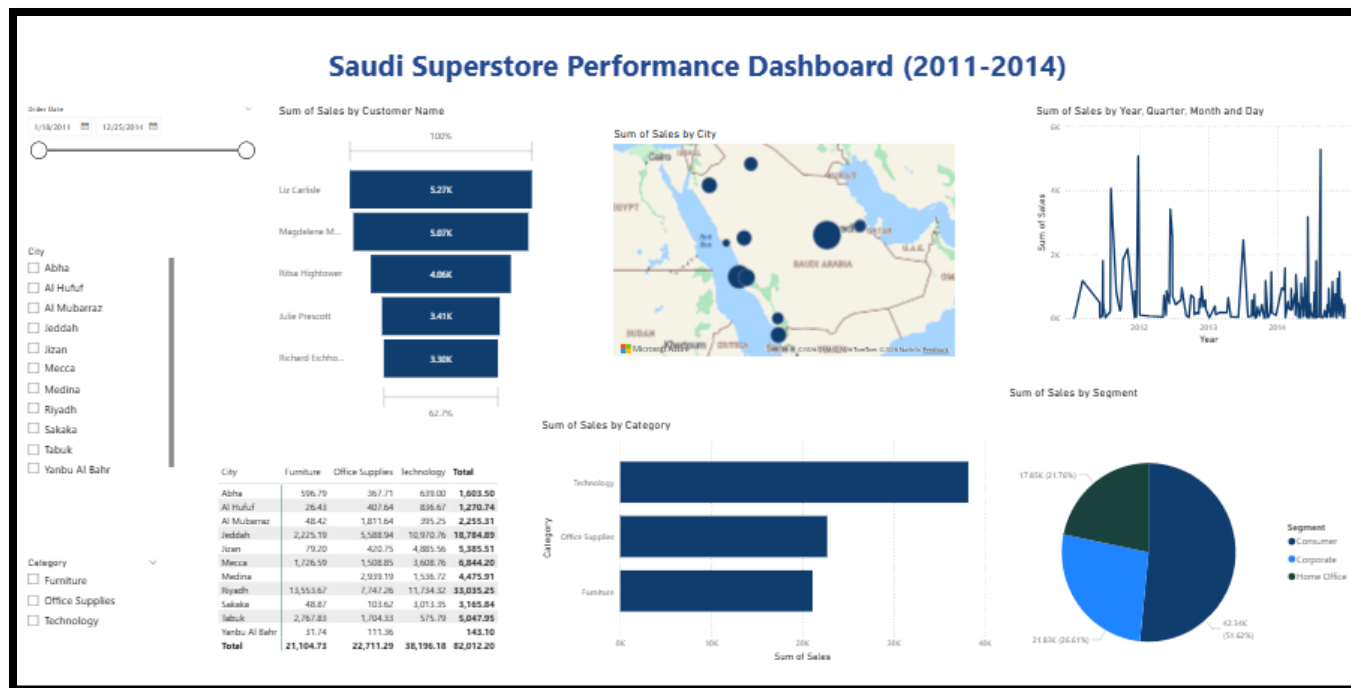


Figure 1: Saudi Superstore Performance Dashboard (2011-2014) — Full View

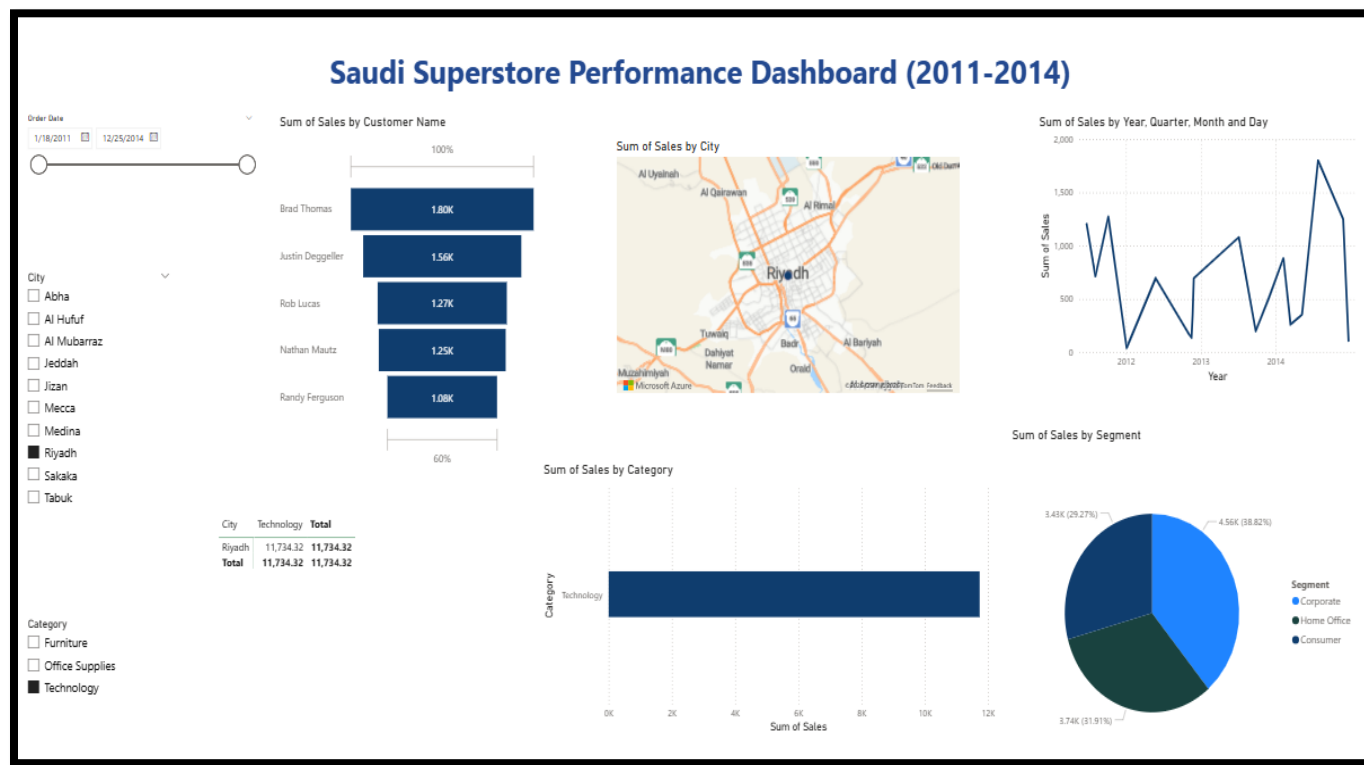


Figure 2: Dashboard filtered by Riyadh — Technology category dominance

Key Insights & Analysis

Sales by Category:

The Technology category is the clear leader in revenue, significantly outperforming Office Supplies and Furniture.

Geographical Performance:

- Riyadh is the top-performing city, with total sales exceeding 33,035 SAR.
- Jeddah follows in second place with approximately 18,784 SAR in sales.
- The Map Visual confirms a high concentration of commercial activity in the Central and Western regions of Saudi Arabia.

Top 5 Customers (Funnel Analysis):

Using the Funnel Chart, we identified the highest-contributing customers. Liz Carlisle leads the group with 5.27K in sales, followed by Magdalene M. (5.07K). This insight is crucial for targeted loyalty programs.

Market Segmentation:

The Consumer segment represents the majority of the market share at 51.62%, followed by the Corporate segment at 26.61%. The Home Office segment is the smallest at 21.76%.

Temporal Trends:

The Line Chart indicates high volatility with significant spikes in late 2014, suggesting strong seasonality in consumer purchasing behavior.

Dashboard Visual Components

1. Line Chart: Visualizes sales fluctuations over the four-year period.
2. Funnel Chart: Provides a streamlined ranking of the Top 5 Customers.
3. Map Visual: Highlights regional sales density across the Kingdom.
4. Matrix Table: Offers a detailed breakdown of sales by City and Category (e.g., showing that Riyadh excels in Furniture with 13,553 SAR).
5. Pie Chart: Illustrates the distribution of sales across different customer segments.
6. Interactive Slicers: Allows users to filter the entire report by Order Date, City, and Category.

Conclusion & Recommendations

The descriptive model reveals that the store's success is heavily driven by the Technology sector and the Riyadh market. To maximize future growth, it is recommended to:

- Strengthen inventory levels for Tech products in the Central Region.
- Develop specialized marketing campaigns targeting the Home Office segment, which currently has the smallest market share (21.76%).
- Investigate the causes of sales dips shown in the Line Chart to ensure more consistent revenue throughout the year.

Part 2: Predictive Analytics (Altair AI Studio)

Predictive Question

Can we predict future sales revenue for retail orders using a Linear Regression model based on historical transactional data, including quantity, shipping cost, product category, customer segment, and discount level?

Tool Justification

Altair AI Studio (formerly RapidMiner Studio) was selected for this predictive model because it is a leading enterprise-grade platform for data science. Its automated machine learning (AutoML) capabilities and visual workflow designer allowed us to build and validate our Linear Regression model with high precision and transparency. The drag-and-drop workflow makes it easy to configure and modify the model pipeline.

Data Description & Preprocessing

Dataset

The dataset used is the Cleaned Saudi Superstore dataset containing 361 records and 23 attributes covering retail orders across Saudi Arabia from 2011 to 2014. The data was pre-cleaned with no missing values, making it ready for model building.

Selected Attributes

Attribute	Type
Sales	Numerical
Quantity	Numerical
Shipping Cost	Numerical
Discount	Numerical
Category	Nominal
Segment	Nominal

Preprocessing Steps

- **Select Attributes:** Only the 6 relevant attributes were retained. Irrelevant columns such as Order ID, Customer Name, City, and Product Name were excluded to reduce noise.
- **Nominal to Numerical:** The categorical attributes Category (Furniture, Office Supplies, Technology) and Segment (Consumer, Corporate, Home Office) were converted to numerical format using Dummy Coding so the Linear Regression model can process them.
- **Set Role:** The Sales attribute was assigned the role of Label (target variable) — the value the model aims to predict.
- **Note:** The Discount column contains all zero values in this dataset. It was retained in the model as it is part of the research question, and this observation was noted in the findings.

Model Configuration & Workflow

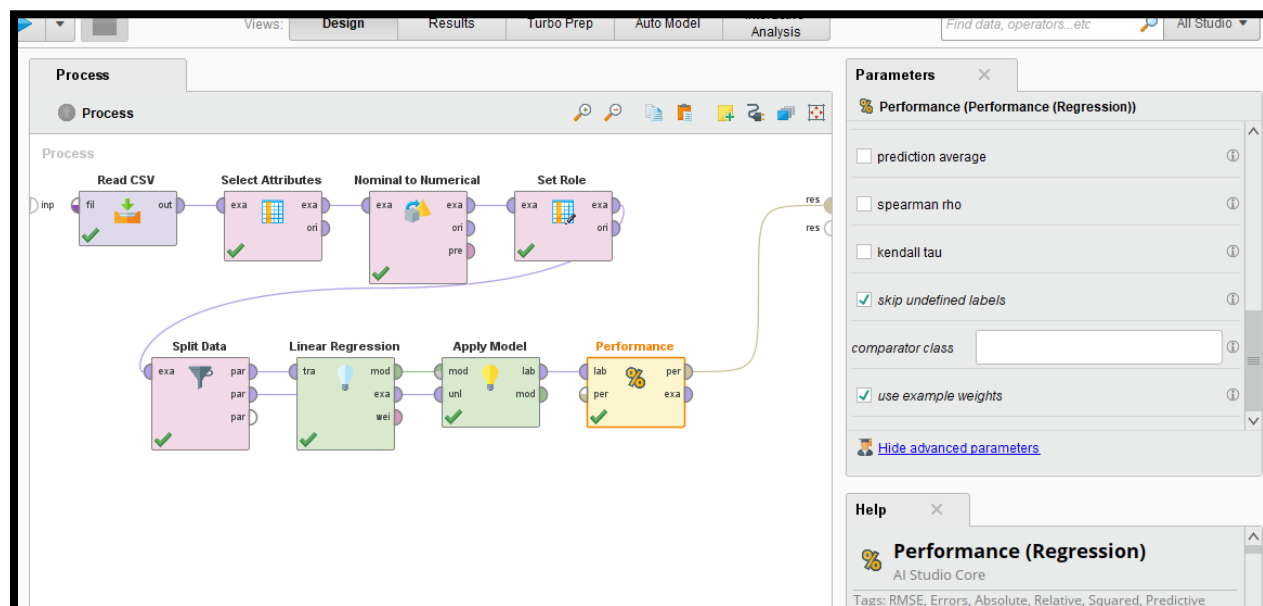


Figure 3: Complete RapidMiner Workflow — Read CSV → Select Attributes → Nominal to Numerical → Set Role → Split Data → Linear Regression → Apply Model → Performance

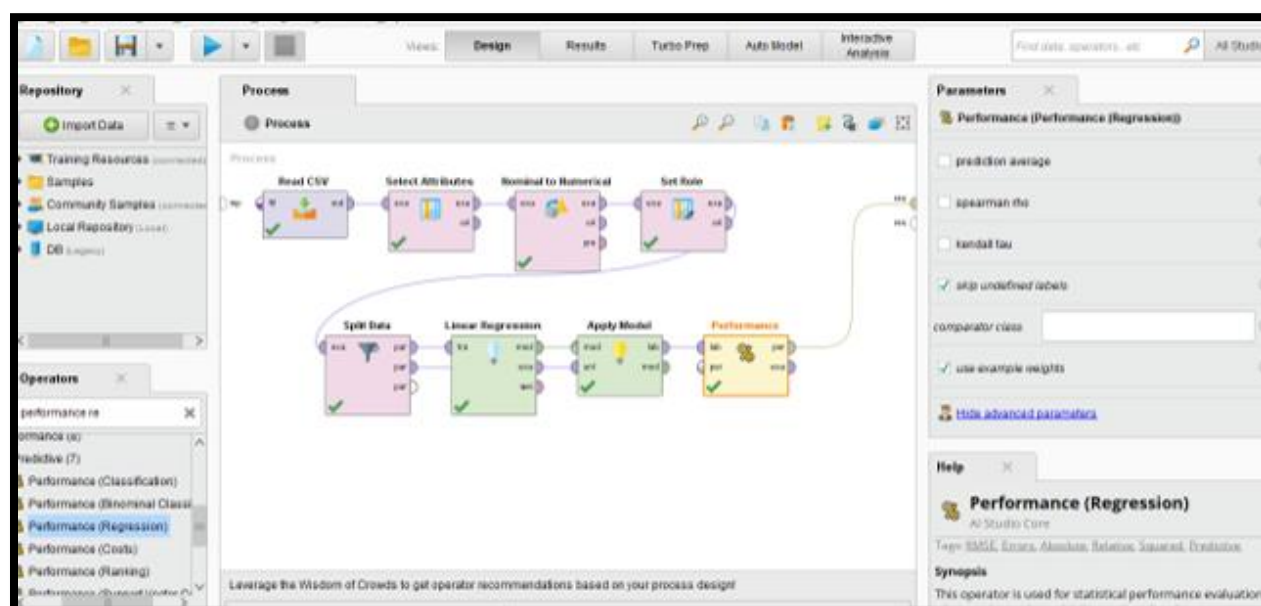


Figure 4: Select Attributes operator — keeping 6 relevant features

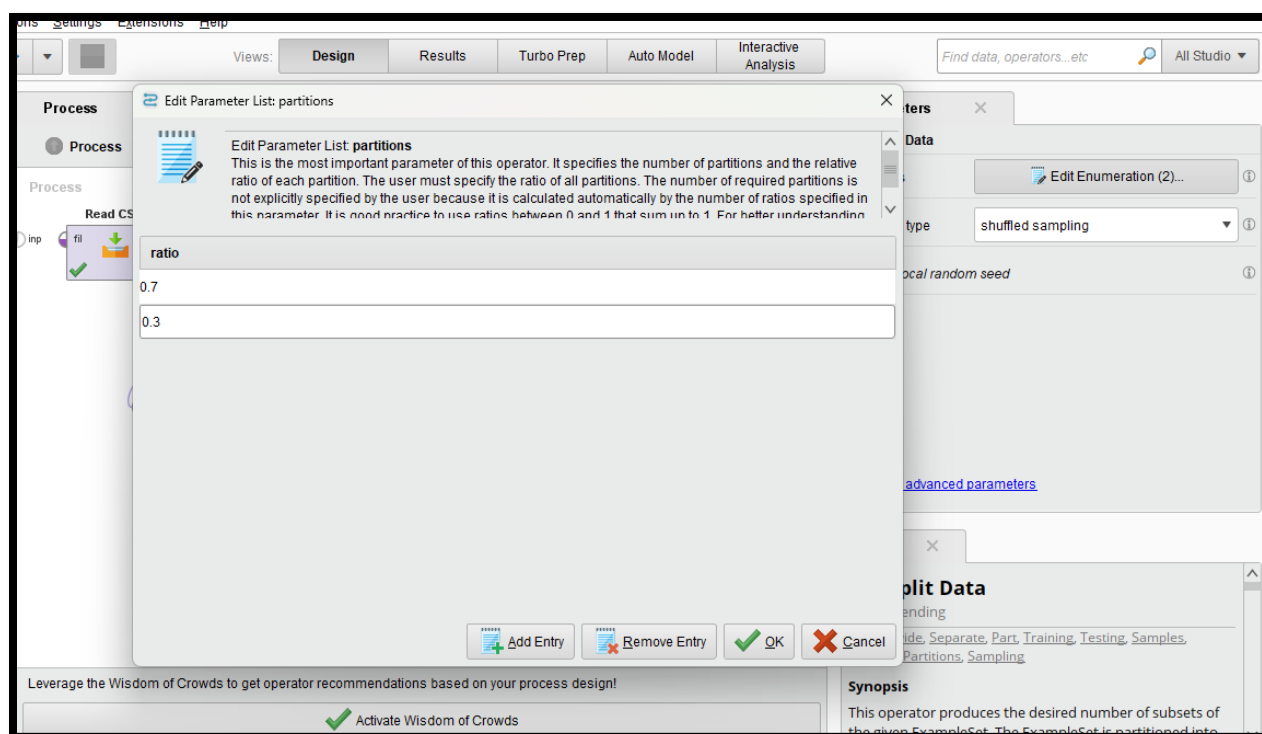


Figure 5: Set Role operator — assigning Sales as the target label

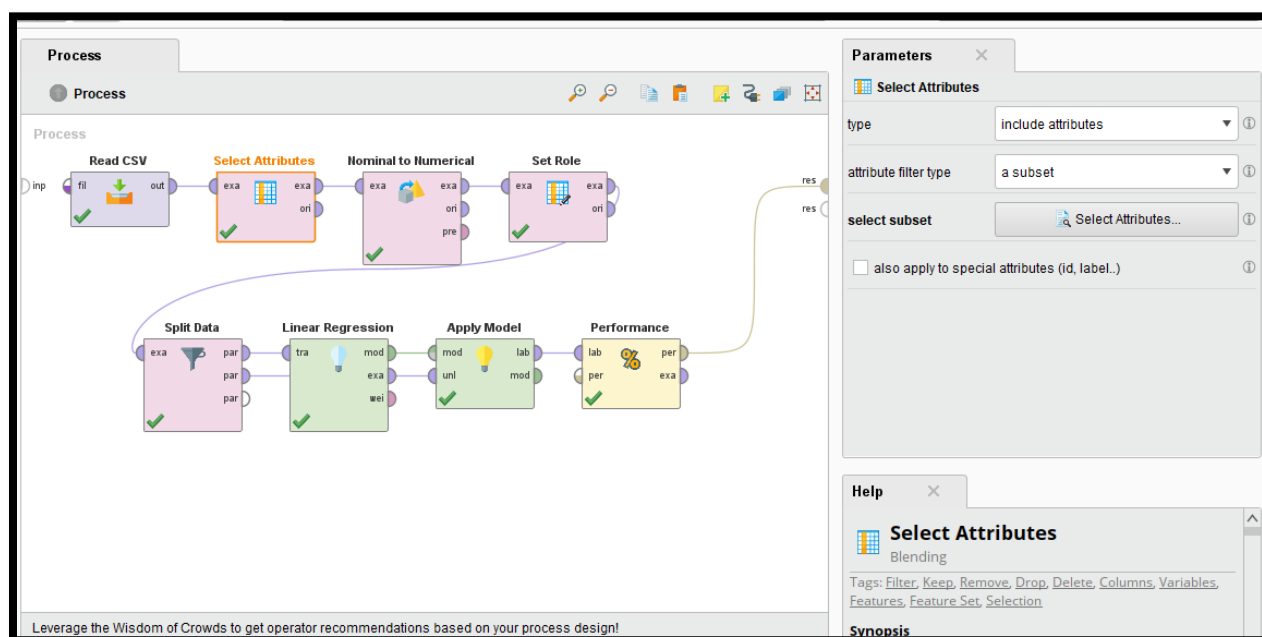
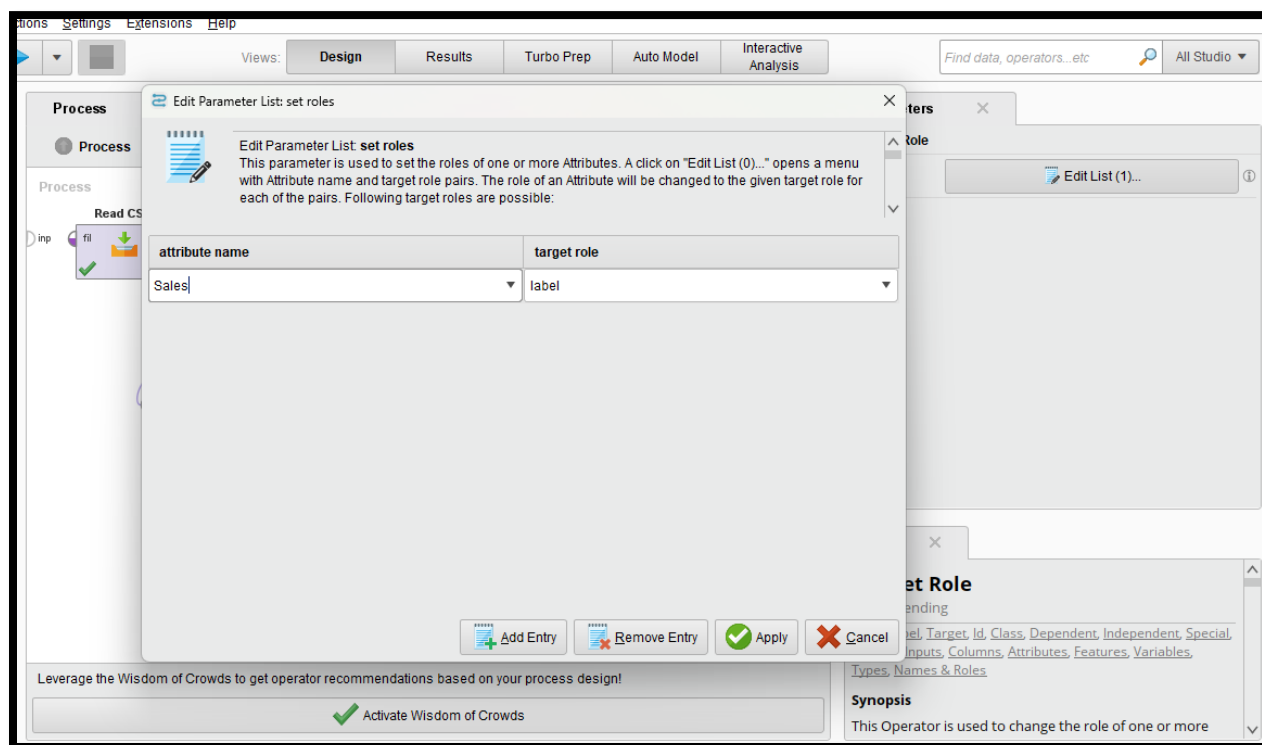


Figure 6: Split Data configuration — 70% training / 30% testing with Shuffled Sampling

Workflow Steps

1. Read CSV – Loads the Cleaned Saudi Superstore CSV file into Altair AI Studio.
2. Select Attributes – Filters the dataset to keep only the 6 selected attributes.
3. Nominal to Numerical – Converts Category and Segment from text to numerical values using Dummy Coding.
4. Set Role – Assigns Sales as the target label for the prediction task.
5. Split Data – Divides the data into 70% Training (approximately 252 records) and 30% Testing (approximately 109 records) using Shuffled Sampling.
6. Linear Regression – Trains the predictive model using the training partition.
7. Apply Model – Applies the trained model to the test partition to generate predictions.
8. Performance (Regression) – Evaluates the model using RMSE and Squared Correlation (R^2).

Performance Metrics Results



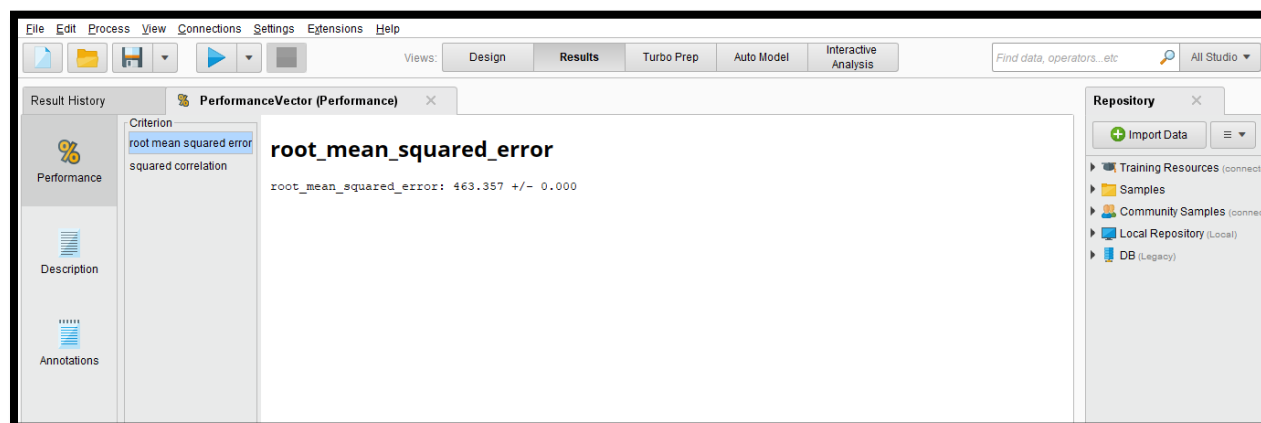


Figure 7: Root Mean Squared Error (RMSE) = 463.357

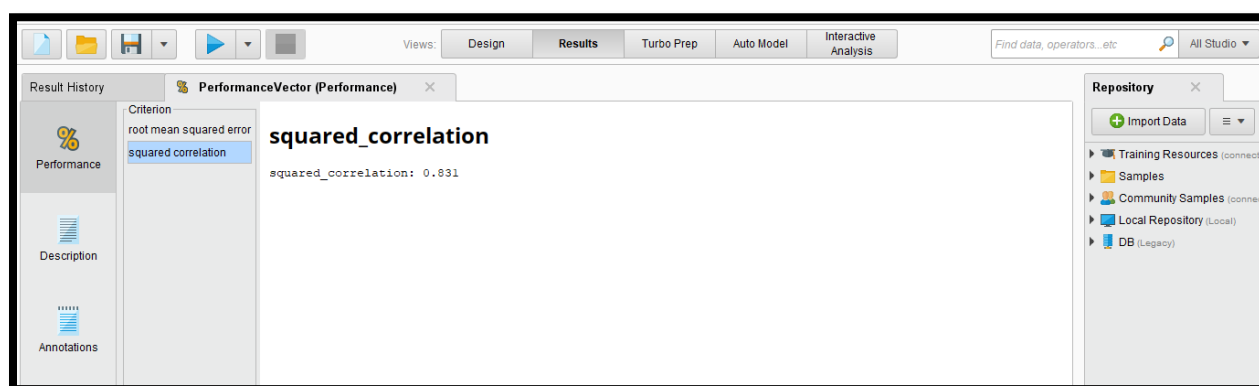


Figure 8: Squared Correlation (R^2) = 0.831

Findings & Observations

Metric	Value
Root Mean Squared Error (RMSE)	463.357
Squared Correlation (R^2)	0.831

$R^2 = 0.831$:

The Linear Regression model explains 83.1% of the variance in sales revenue. This is considered strong predictive performance, indicating that the selected features are effective predictors of Sales.

RMSE = 463.357:

The model has an average prediction error of approximately 463 SAR. This is acceptable given the wide range of sales values in the dataset, which span from a minimum of 3.87 SAR to a maximum of 5,211.12 SAR.

Discount Observation:

The Discount attribute contains all zero values in this dataset and therefore did not contribute to the model's predictive power.

Conclusion

Overall, the Linear Regression model demonstrates strong predictive capability with an R^2 of 0.831. The model successfully identifies the relationship between the selected input features and sales revenue. Future improvements could include collecting a larger dataset, adding more relevant features, or applying more advanced algorithms such as Random Forest to further reduce the prediction error.

Part 3: Prescriptive Analytics & Strategic Integration

Overview & Objective

Prescriptive analytics goes beyond describing what happened (descriptive) and predicting what will happen (predictive) it answers the critical question: What should we do? This model synthesizes findings from Raghad's Power BI dashboard and Fawziyah's Altair AI Studio regression model to generate evidence-based business recommendations for the Saudi Superstore management team.

Prescriptive Question

"Which product categories, sub-categories, and cities should be prioritized for marketing investment and resource allocation to maximize profit, and what strategies can be used to optimize shipping cost and improve overall profitability?"

Tool Justification

The prescriptive model in this section applies a structured Decision Framework methodology, synthesizing quantitative outputs from Power BI (visual analytics) and Altair AI Studio (statistical modeling). This approach is appropriate for prescriptive analytics because it requires domain judgment, contextual reasoning, and the ability to weigh multiple business constraints — tasks that benefit from human interpretation of quantitative outputs. This reflects the Low-Code / No-Code BI philosophy introduced in Assignment 1, where insights are extracted and acted upon without requiring complex programming.

Configuration & Setup (Rule-Based Decision Model)

The prescriptive model was built as a Rule-Based Decision Tree that translates the outputs of the descriptive and predictive models into automated business rules. The logic follows four sequential decision points derived directly from the data findings of Part 1 (Power BI) and Part 2 (Altair AI Studio). Each decision node applies an IF-THEN condition, branching into a prescriptive action based on whether the condition is met.

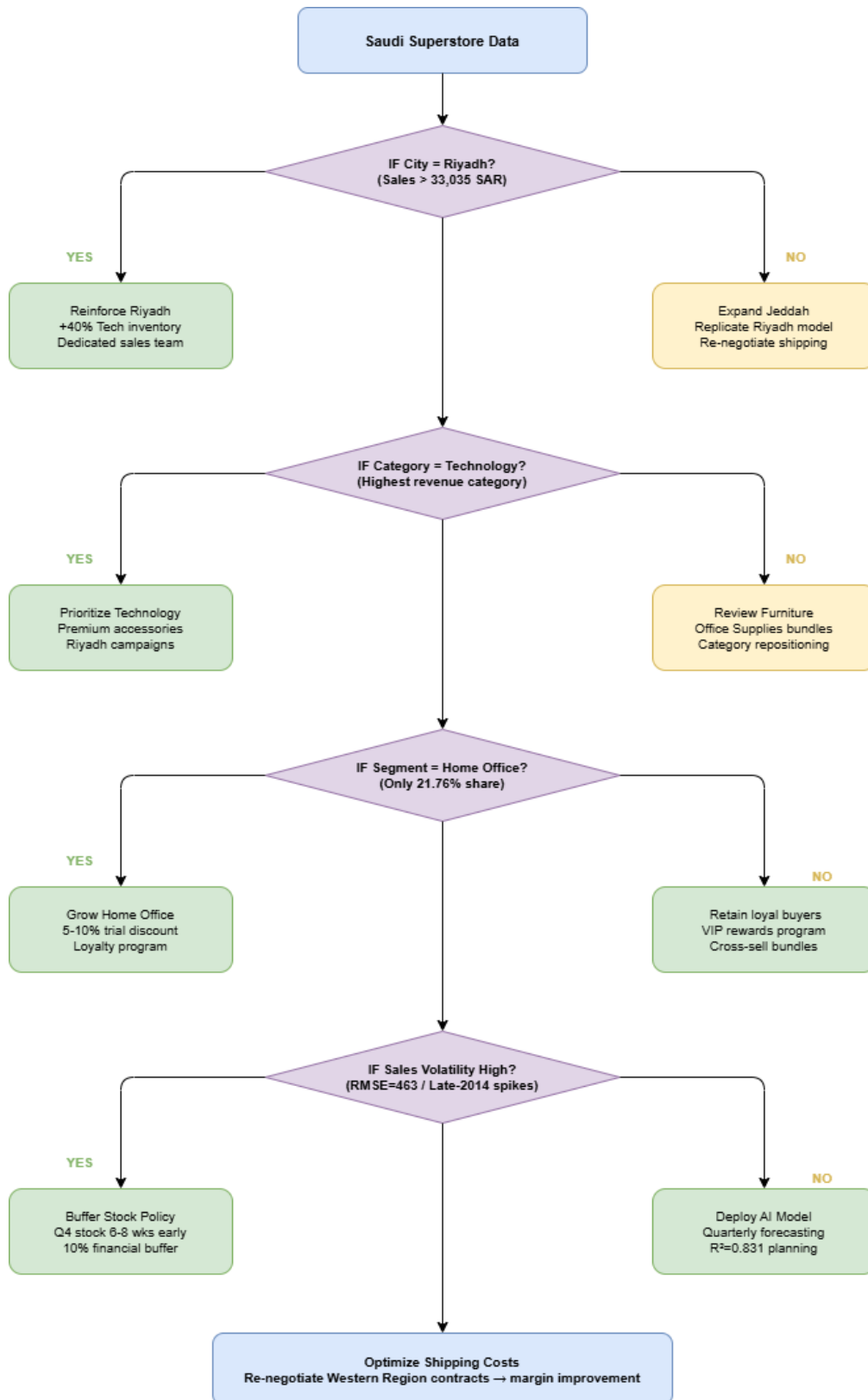


Figure 9: Rule-Based Prescriptive Decision Tree (Saudi Superstore (2011-2014))

The decision tree applies the following four IF-THEN business rules derived from the data:

- IF City = Riyadh (Sales > 33,035 SAR) → Increase Technology inventory by 40% and assign dedicated sales team.
- IF Category = Technology (top revenue driver) → Launch premium accessory line and targeted marketing campaigns in Central Region.
- IF Segment = Home Office (share < 25%) → Apply 5-10% trial discount and introduce loyalty program to grow segment share to 30%.
- IF Sales Volatility High (RMSE > 400 SAR or late-2014 spikes detected) → Activate Q4 buffer stock policy 6-8 weeks early and maintain 10% financial safety margin.

Business Questions Addressed

- Which product categories and cities should the business prioritize to maximize profit?
- How should marketing budgets be reallocated based on customer segment performance?
- What actions should be taken in response to predicted sales growth?
- How can shipping costs be optimized to improve overall profitability?
- What strategies should be implemented to address the underperformance of the Home Office segment?

Summary of Input Findings

The prescriptive recommendations below are grounded in the following quantitative findings from the descriptive and predictive models:

Source Model	Key Finding	Business Implication
Power BI (Descriptive)	Riyadh: 33,035 SAR in sales	Central region dominates revenue
Power BI (Descriptive)	Jeddah: 18,784 SAR	Strong second market
Power BI (Descriptive)	Technology = highest category	Tech is the primary growth driver
Power BI (Descriptive)	Consumer segment = 51.62%	Dominant customer base
Power BI (Descriptive)	Home Office = only 21.76%	Significant untapped opportunity
Power BI (Descriptive)	Sales spikes in late 2014	Strong Q4 seasonality
AI Studio (Predictive)	$R^2 = 0.831$	Sales are 83.1% predictable
AI Studio (Predictive)	RMSE = 463.357 SAR	Reliable for planning purposes
AI Studio (Predictive)	Discount = all zeros	Untested pricing lever available

Prescriptive Recommendations

Recommendation 1: Reinforce the Riyadh-Technology Growth Engine

The data unambiguously identifies Riyadh and Technology as the dual pillars of store performance. Riyadh accounts for a disproportionate share of total revenue, and Technology drives the highest category-level sales across the board.

- Increase Technology product inventory in Riyadh warehouses by at least 40% ahead of Q4 to prevent stockouts during peak demand periods.
- Assign a dedicated sales team to corporate clients in Riyadh who purchase Technology products, as this segment likely drives high-value, repeat orders.
- Develop Riyadh-specific promotional campaigns targeting both Consumer and Corporate segments for Technology products during high-sales months.
- Expand the Riyadh product range to include premium Technology accessories to increase average order value.

Recommendation 2: Accelerate Growth in Jeddah

Jeddah represents a significant and growing second market. The Map Visual shows high density in Jeddah, and the predictive model highlights Shipping Costs as a key factor — making logistics optimization in the Western Region a strategic priority.

- Replicate the Riyadh commercial model in Jeddah — prioritize Technology and Office Supplies stocking.
- Re-negotiate contracts with Western Region carriers to lower shipping costs and increase net profit from Jeddah orders.
- Conduct targeted outreach campaigns in cities with below-average performance to increase brand awareness.

Recommendation 3: Grow the Home Office Segment

The Home Office segment currently represents only 21.76% of the market — the smallest of the three customer segments. A Trial Discount Phase of 5-10% is prescribed to capture more market share while monitoring the impact on profit margins.

- Design a dedicated Home Office product bundle combining Office Supplies and Technology items at a preferential price point.
- Launch a targeted digital marketing campaign highlighting products suited to home office environments.
- Introduce a loyalty program for Home Office customers with a target of growing segment share to 30%.
- Test a 5-10% discount on selected Home Office products to stimulate demand while monitoring margin impact.

Recommendation 4: Address Seasonal Sales Volatility

The Power BI Line Chart reveals significant spikes in late 2014, indicating strong seasonality. Without proactive planning, this volatility can lead to stockouts during peaks and excess inventory during troughs.

- Maintain a 10% safety buffer in financial planning to account for the average prediction error of 463 SAR (RMSE from the predictive model).
- Implement a quarterly demand forecasting cycle using the AI Studio model ($R^2 = 0.831$) to generate sales estimates for each upcoming quarter.
- Establish a buffer stock policy for Technology and Furniture products 6-8 weeks before anticipated Q4 demand peaks.
- During low-sales periods, launch clearance promotions or bundle deals to stabilize revenue and reduce holding costs.

Recommendation 5: Operationalize the Predictive Model

The AI Studio model achieves an R^2 of 0.831, explaining over 83% of sales variance. This level of accuracy is sufficient to support practical business planning.

- Deploy the AI Studio model as a quarterly sales planning tool: input expected order volumes by category and segment to generate revenue forecasts.
- Use the model's coefficients to identify which features (Quantity, Category) have the greatest impact on Sales as key performance indicators.
- Explore introducing a structured discount policy through a controlled A/B test to validate whether discounting drives incremental volume.
- Consider adding features such as seasonality flags and regional indicators to further reduce the RMSE below 463.357 SAR.

Prescriptive Decision Matrix

Business Decision Area	Priority	Expected Impact	Key Action
Technology & Riyadh focus	High	Revenue growth 20-40%	Increase inventory + dedicated sales team
Jeddah market expansion	High	Market share growth	Replicate Riyadh model + logistics renegotiation
Home Office segment growth	Medium	Segment share 21% → 30%	Bundle products + 5-10% trial discount
Seasonal demand planning	High	Reduce stockouts + volatility	Q4 buffer stock + 10% financial safety margin
Predictive model deployment	Medium	Improved planning accuracy	Quarterly AI Studio forecast cycles
Shipping cost optimization	Medium	Margin improvement	Re-negotiate Western Region contracts

Project Integration: Connecting All Three Models

The three analytical models form a coherent, end-to-end Business Intelligence pipeline that transforms raw transactional data into actionable strategic guidance:

Layer	Model	Tool	Contribution
Layer 1	Descriptive	Power BI	Reveals the current state: who bought what, where, and when. Establishes baseline performance for Riyadh, Technology, and Consumer segments.
Layer 2	Predictive	Altair AI Studio	Quantifies relationships between inputs and Sales. Confirms future sales can be forecast with 83.1% accuracy ($R^2 = 0.831$).
Layer 3	Prescriptive	Decision Framework	Synthesizes Layers 1 & 2 to answer: Given what we know and expect, what specific actions should management take?

Key Cross-Model Insights

- Riyadh dominance in the descriptive model combined with the strong R^2 in the predictive model confirms that a Riyadh-focused strategy can be planned with high confidence.

- The Home Office segment's small size in the descriptive model represents a gap that can be quantified using the predictive model by simulating Segment = Home Office scenarios.
- The all-zero discount column (flagged in the predictive model) reveals that the business has never used discounts as a sales lever — a significant untested strategic tool.
- The high RMSE relative to order size is consistent with the sharp sales spikes in the Power BI Line Chart, reinforcing the recommendation for Q4 buffer stock policies.

Conclusion

This BI solution provides a full-spectrum view of the Saudi Superstore. We started by identifying historical winners in Riyadh and the Consumer segment using Power BI. We then validated our ability to predict future revenue with 83.1% accuracy using Altair AI Studio. Finally, these insights lead to a prescriptive strategy: Focus on Technology, invest in Riyadh and Jeddah, reward top customers, and cautiously introduce discounts to grow the Home Office segment.

The integration of all three models demonstrates that Business Intelligence is most powerful not as isolated tools, but as a coordinated pipeline from understanding the past, to predicting the future, to prescribing the right actions for sustained growth.

References

- **Dataset Source:**

- Kaggle. (2024). *Global Superstore Dataset: Saudi Arabia Retail Subset*. Retrieved from <https://www.kaggle.com/datasets>.

- **Software & Tools:**

- Microsoft. (2026). *Power BI Desktop: Interactive Data Visualization Software*. Retrieved from <https://powerbi.microsoft.com>.
- Altair Engineering Inc. (2026). *Altair AI Studio: The Next Generation of RapidMiner for Enterprise Data Science*. Retrieved from <https://altair.com/ai-studio>.

- **Courses & Reserches:**

- Coursera. (2025). *Top Business Intelligence Trends in 2025*. Coursera.
- Prometai. (2025). *Trends in AI BI & Conversational Analytics*. Prometai Research.